

Name of Experiment: A Program to Extract all Identifiers from a full Source Code.

Introduction: The main objective of this program is to perform **Lexical Analysis** by identifying **Identifiers** from a given source code written in a C-like language.

In compiler design, identifiers are user-defined names such as variable names, function names, or object names. The program reads source code from the user, separates words from the code, removes reserved keywords, and finally displays only the valid identifiers.

The input process continues until the user enters *END*

Code:

```
import re

keywords = {
    "int", "float", "double", "char", "if", "else", "while",
    "for", "return", "void", "break", "continue", "switch",
    "case", "default", "do", "struct", "typedef", "long",
    "short", "unsigned", "signed", "const", "static"
}

print("Enter Your Source Code: ")

lines = []
while True:
    line = input()
    if line == "END":
        break
    lines.append(line)

code = "\n".join(lines)

words = re.findall(r'[A-Za-z_][A-Za-z0-9_]*', code)

identifiers = set()

for word in words:
    if word not in keywords:
        identifiers.add(word)

print("\nIdentifiers found:")
for identifier in identifiers:
    print(identifier)
```

Input:

```
int age = 10;
float salary;
return 0;
}

END
```

Output:

```
Identifiers found:
age
salary
```

Discussion: The program correctly identified *age* and *salary* as valid identifiers from the given source code. Reserved keywords such as *int*, *float*, and *return* were ignored because they are predefined language keywords. Numeric values and symbols were also excluded since they are not identifiers. This output shows that the program successfully performs basic lexical analysis by separating identifiers from keywords and other tokens.